

BrightCart Retail Ltd. – 2025 Business Performance Analysis

Simulated Business Analysis Project I Microsoft Excel

Project Overview

This independent simulated business analysis project used Microsoft Excel to analyse BrightCart Retail Ltd.'s 2025 sales performance. The project involved cleaning and validating 2,000 raw transaction records, analysing 1,985 valid orders, evaluating revenue, profitability, customer satisfaction, delivery performance and product returns, and translating the results into evidence-based recommendations for management.

The purpose of the project was not simply to calculate business metrics, but to demonstrate how structured data analysis can be used to identify performance issues, investigate potential causes and support management decision-making.

Business Problem

BrightCart Retail Ltd. needed a clearer understanding of its overall business performance during 2025. Although sales transactions were available, management required analysis to determine which areas of the business were performing well, where profitability was weaker, which products and regions were associated with higher returns or lower customer satisfaction, and whether delivery performance was affecting the customer experience.

The raw dataset also contained several data-quality issues that needed to be addressed before reliable analysis could be performed. These included duplicate transactions, missing values, inconsistent text formatting and dates stored in different formats.

The analysis therefore focused on turning the raw transactional data into reliable management information that could support practical business decisions.

Project Objectives

The main objectives of the analysis were to:

- Clean and prepare the raw transactional dataset for analysis.
- Measure overall revenue, profit and profit margin.
- Analyse monthly sales and profitability trends.
- Compare performance across regions.
- Compare revenue and profitability across product categories and individual products.
- Assess customer-segment performance.
- Analyse product and regional return rates.
- Investigate the relationship between delivery time and customer satisfaction.
- Identify areas associated with lower customer satisfaction.
- Develop practical recommendations based on the findings.
- Create an Excel dashboard that communicates the most important results clearly to management.

Dataset

The project used a simulated retail sales dataset containing **2,000 transaction records** covering the 2025 calendar year.

The dataset included information such as:

- Order ID
- Order date
- Region
- Customer type
- Product category
- Product
- Quantity
- Unit price
- Unit cost
- Discount percentage
- Delivery days
- Customer satisfaction score
- Return status
- Sales channel
- Payment method

The dataset was deliberately designed to reflect the type of data-quality and business-performance issues that may be encountered in a real organisational environment.

Data Cleaning and Preparation

Before performing the analysis, I reviewed the dataset to identify quality issues that could affect the reliability of the results.

Several issues were identified, including:

- **15 duplicated orders**, resulting in 30 rows being flagged during the duplicate check.
- **24 missing customer satisfaction scores.**
- **6 missing customer-type values.**
- Inconsistent regional naming and spacing.
- Inconsistent product-category formatting.
- Extra spaces within product names.
- Dates stored in different formats, including text-formatted dates.

After removing the duplicate transactions, the final dataset contained **1,985 valid orders.**

Rather than modifying the original raw data directly, I created cleaned and calculated fields so that the original information remained available for reference.

Examples of the data-cleaning activities included:

- Standardising customer types using Excel text-cleaning functions.
- Assigning **“Unknown”** to missing customer-type values.
- Standardising region and product-category names.
- Correcting inconsistent values such as variations of **Midlands.**
- Converting text-formatted dates into valid Excel dates.
- Removing unnecessary spaces from product names.

- Leaving missing satisfaction scores blank rather than replacing them with artificial values.

This approach helped preserve data integrity while creating a reliable dataset for analysis.

Calculated Business Metrics

Additional calculated fields were created to support the analysis.

Revenue

Revenue was calculated using:

$$\text{Quantity} \times \text{Unit Price} \times (1 - \text{Discount Percentage})$$

Total Cost

Total cost was calculated using:

$$\text{Quantity} \times \text{Unit Cost}$$

Profit

Profit was calculated as:

$$\text{Revenue} - \text{Total Cost}$$

Profit Margin

Profit margin was calculated as:

$$\text{Profit} \div \text{Revenue}$$

A monthly field was also created from the cleaned order date to support monthly trend analysis.

Analysis and Key Findings

1. Overall Business Performance

BrightCart processed **1,985 valid orders** during 2025.

The company generated:

- **Revenue:** £331,552.88
- **Profit:** £123,547.26
- **Overall Profit Margin:** 37.26%

These figures provide a baseline against which more detailed regional, product and operational performance can be assessed.

2. Monthly Revenue and Profit Trends

Business performance generally strengthened towards the end of the year.

December recorded the strongest performance:

- **Revenue:** £43,503.26
- **Profit:** £14,783.08

February recorded the lowest:

- **Revenue:** £18,837.46
- **Profit:** £7,185.14

There was particularly strong revenue growth during **September, November and December**, suggesting a significant year-end seasonal pattern.

However, increased revenue did not always result in a proportional increase in profitability, indicating that factors such as product mix, discounts or costs may also influence monthly performance.

Management implication

BrightCart should plan stock levels, staffing and operational capacity around the stronger year-end period while continuing to monitor whether higher sales volumes are translating into equally strong profit performance.

3. Regional Performance

Regional analysis revealed differences between sales performance and profitability.

London

- Revenue: **£94,673.78**
- Profit: **£36,002.34**
- Profit Margin: **38.03%**

London generated the highest revenue and total profit.

South

- Revenue: **£94,091.64**
- Profit: **£32,935.14**
- Profit Margin: **35.00%**

South generated almost as much revenue as London but had the **lowest regional profit margin**.

Midlands

- Revenue: **£71,983.33**
- Profit: **£27,769.32**
- Profit Margin: **38.58%**

Although Midlands generated less revenue than London and South, it achieved the **highest regional profit margin**.

North

- Revenue: **£70,804.12**
- Profit: **£26,840.45**
- Profit Margin: **37.91%**

Management implication

The South region requires further investigation because strong sales are not translating into equally strong profitability.

4. Product Category Performance

The analysis demonstrated that high revenue does not necessarily mean high profitability.

Furniture

- Revenue: **£102,600.00**
- Profit Margin: **33.27%**

Furniture generated the highest category revenue but recorded the **lowest category profit margin**.

Electronics

- Revenue: **£101,538.30**
- Profit: **£36,267.85**
- Profit Margin: **35.72%**

Electronics generated the highest total category profit.

Home & Kitchen

- Revenue: **£82,171.12**
- Profit Margin: **39.35%**

Office

- Revenue: **£45,243.49**
- Profit Margin: **46.00%**

Office generated the lowest revenue but achieved the **highest category profit margin**.

Management implication

Management should review the pricing and cost structure of Furniture products while protecting the strong margins achieved within the Office category.

5. Individual Product Profitability

Product-level analysis identified several items generating strong revenue but comparatively weaker margins.

The clearest example was **SmartHome Hub**:

- Revenue: **£40,967.26**
- Profit: **£11,995.60**
- Profit Margin: **29.28%**

It generated the highest individual-product revenue but achieved a margin considerably below the company-wide margin of 37.26%.

Other products showing relatively strong revenue, but weaker margins included:

Storage Cabinet

- Revenue: £36,266.77
- Profit Margin: 32.38%

Office Chair

- Revenue: £28,326.44
- Profit Margin: 32.34%

Management implication

Management should examine product costs, discounting and pricing for these products to understand why strong sales are producing comparatively weaker margins.

6. Customer Segment Performance

Individual customers contributed the largest proportion of revenue and profit.

Individual

- Revenue: **£164,250.70**
- Profit: **£61,819.15**
- Profit Margin: **37.64%**

Small Business

- Revenue: **£105,611.60**
- Profit: **£40,070.69**
- Profit Margin: **37.94%**

Corporate

- Revenue: **£61,063.94**
- Profit: **£21,410.14**
- Profit Margin: **35.06%**

Small Business customers produced a slightly stronger profit margin than Individual customers, while corporate customers recorded the lowest margin among the main identified customer groups.

Management implication

BrightCart should continue supporting its Individual and Small Business customer base while investigating opportunities to improve profitability within the corporate segment.

Operational and Customer Experience Analysis

7. Product and Regional Return Rates

The **South region** recorded the highest regional return rate:

- South: **11.76%**
- North: 7.98%
- Midlands: 7.78%
- London: 7.51%

At category level:

- Electronics: **11.69%**
- Office: 8.60%
- Home & Kitchen: 7.17%
- Furniture: 6.95%

At individual-product level, several products stood out:

- **SmartHome Hub: 18.75%**
- **Desk Lamp: 15.09%**
- Office Chair: 10.34%
- Wireless Headset: 10.32%

SmartHome Hub therefore represented a particularly important concern because it combined **strong sales with a high return rate and comparatively weak profitability.**

Management implication

BrightCart should investigate return reasons, product quality, supplier performance and product information for SmartHome Hub and other high-return products.

8. Delivery Performance and Customer Satisfaction

The analysis identified a clear relationship between longer delivery times and lower customer satisfaction.

Average satisfaction remained relatively strong for orders delivered within five days:

- Day 1: 4.15
- Day 2: 4.27
- Day 3: 4.24
- Day 4: 4.23
- Day 5: 4.24

From six days onwards, satisfaction declined significantly:

- Day 6: **3.68**
- Day 7: 3.61
- Day 8: **3.19**
- Day 9: 3.10
- Day 10: **3.00**
- Day 11: **3.00**

The results suggest that **five days represents an important service threshold.** Deliveries extending beyond this point were associated with substantially lower customer satisfaction.

Management implication

Where operationally possible, BrightCart should aim to complete deliveries within five days.

9. Sources of Customer Dissatisfaction

Further analysis was conducted to identify where lower customer satisfaction was concentrated.

Regional satisfaction

- North: 4.22
- South: **4.07**
- Midlands: 4.16
- London: 4.18

South had the lowest regional satisfaction score.

Category satisfaction

- Electronics: 4.10
- Office: 4.26
- Home & Kitchen: 4.23
- Furniture: **3.99**

Furniture had the lowest category satisfaction score.

Products with lower satisfaction

- **SmartHome Hub: 3.76**
- Side Table: 3.94
- Office Chair: 3.96
- Bookshelf: 3.99

SmartHome Hub again emerged as the most significant individual-product issue.

Key Management Recommendations

Based on the combined analysis, I identified three priority areas for management.

1. Improve delivery performance

BrightCart should aim to keep deliveries within five days wherever operationally possible. Customer satisfaction remained above 4.15 for deliveries completed within one to five days but declined sharply once delivery exceeded five days.

Management could review delivery partners, order-processing time, fulfilment procedures and regional logistics to identify the causes of longer delivery times.

2. Conduct a root-cause review of high-return products

SmartHome Hub should receive particular attention because it recorded:

- Highest product return rate: **18.75%**
- Lowest product satisfaction score: **3.76**
- Profit margin: **29.28%**

Management should investigate return reasons, product quality, supplier performance, customer expectations, product descriptions and discounting before deciding on corrective action.

Desk Lamp and other relatively high-return products should also be monitored.

3. Investigate South regional performance

The South region demonstrated an unusual combination of:

- Revenue of **£94,091.64**
- Highest regional return rate: **11.76%**

- Lowest regional satisfaction score: **4.07**
- Lowest regional profit margin: **35.00%**

Despite generating almost as much revenue as London, South underperformed on several quality and profitability measures.

Management should investigate regional pricing, discount behaviour, delivery performance, product mix and return patterns to determine the underlying causes.

Excel Dashboard

I developed a management dashboard in Microsoft Excel to communicate the most important findings clearly.

The dashboard includes four headline KPIs:

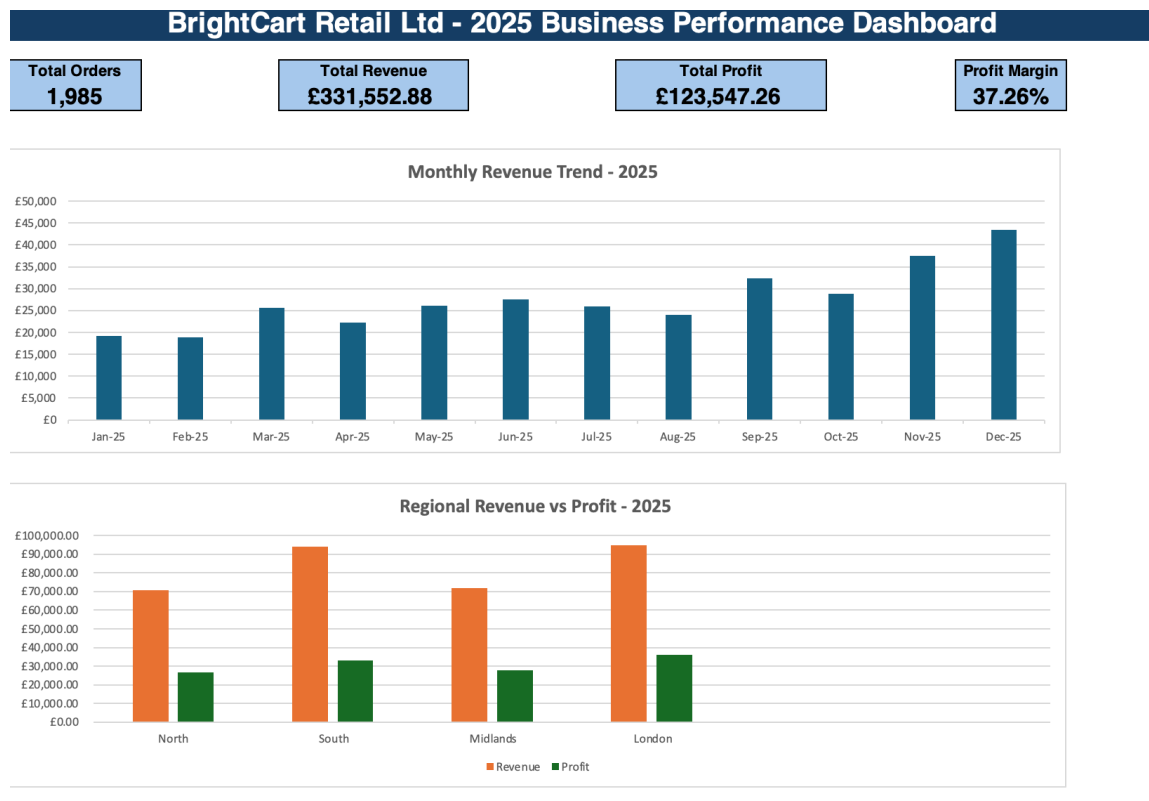
- **Total Orders – 1,985**
- **Total Revenue – £331,552.88**
- **Total Profit – £123,547.26**
- **Profit Margin – 37.26%**

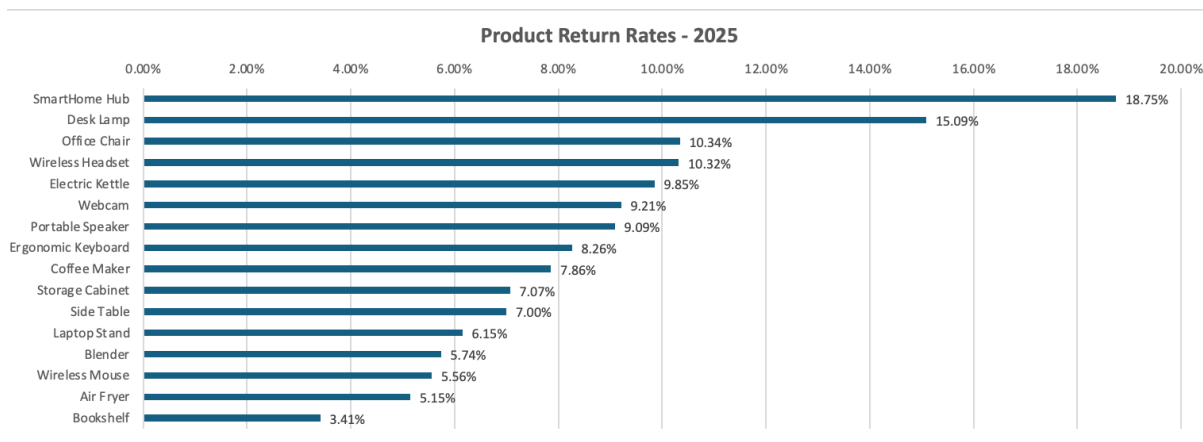
The dashboard also contains visualisations showing:

- Monthly Revenue Trend
- Regional Revenue versus Profit
- Delivery Time versus Customer Satisfaction
- Product Return Rates

The product return-rate chart was ranked from highest to lowest so that management can immediately identify products requiring attention.

The dashboard was designed to provide management with a concise view of both financial performance and customer-experience issues.





Tools and Excel Techniques Used

The project was completed using **Microsoft Excel**.

Techniques used included:

- Data cleaning and validation
- IF
- TRIM
- PROPER
- DATEVALUE
- Duplicate checking
- Calculated columns
- Revenue calculations
- Cost calculations
- Profit and profit-margin calculations
- SUBTOTAL
- Filtering
- Percentage calculations
- Average calculations
- Sorting and ranking
- Business KPI analysis

- Column charts
- Bar charts
- Line charts
- Dashboard development
- Business interpretation and recommendation development

Skills Demonstrated

This project demonstrates my ability to:

- Translate raw business data into meaningful information.
- Identify and resolve common data-quality issues.
- Calculate and interpret financial and operational KPIs.
- Analyse trends across time, products, regions and customer segments.
- Identify patterns requiring further business investigation.
- Distinguish between strong sales performance and strong profitability.
- Analyse customer satisfaction and operational performance together.
- Develop evidence-based management recommendations.
- Design clear Excel dashboards.
- Communicate analytical findings in a business-focused manner rather than presenting numbers without context.

Project Outcome

The analysis showed that BrightCart was profitable overall, but that performance varied considerably across products, regions and operational measures.

The most important insight was that **strong revenue alone did not always indicate strong business performance**.

For example, the South region generated substantial revenue but also recorded the highest return rate, lowest regional satisfaction and weakest regional profit margin. Similarly, SmartHome Hub generated the highest individual-product revenue but also had the highest return rate, lowest product satisfaction and a relatively weak profit margin.

The analysis therefore demonstrated the importance of considering **revenue, profitability, customer experience and operational performance together** when making business decisions.

Limitations

This project uses a **simulated dataset** and therefore does not represent the actual performance of a real company.

The analysis is also based on one year of transactional data. Additional information such as customer demographics, detailed return reasons, supplier performance, inventory availability, marketing expenditure and individual delivery-provider performance would support deeper root-cause analysis.

The findings therefore identify areas requiring management attention but should not automatically be interpreted as proof of causation.

Conclusion

This project demonstrates an end-to-end business analysis process, beginning with raw and imperfect transactional data and progressing through data cleaning, validation, KPI calculation, exploratory analysis, business interpretation, dashboard development and management recommendations.

The analysis identified meaningful relationships between sales, profitability, product returns, delivery performance and customer satisfaction and converted those findings into practical areas for further management action.

It demonstrates how Microsoft Excel can be used not only for calculations, but as a practical business analysis tool for supporting evidence-based decision-making.